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RESEARCH ARTICLE

Relay Selection Algorithm for Air–Ground Cooperative Communication Based on System Capacity and Joint User Equalization

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ABSTRACT Cooperative communication networks employing Uncrewed Aerial Vehicles (UAVs) as aerial relays can significantly enhance user communication quality and extend transmission range. Within such networks, an effective relay selection strategy is crucial for improving the user data rate. To balance system capacity with user fairness, this paper proposes a Scatter Search Relay Selection (SSRS) algorithm based on a network formation game model. The objective is to identify the optimal relay selection strategy for a cooperative communication system involving multiple UAV relays and multiple user pairs. The effectiveness and scalability of the proposed SSRS algorithm are evaluated through simulation experiments. Results demonstrate that SSRS outperforms other mainstream algorithms in terms of both system capacity and user fairness. In summary, the proposed algorithm provides a viable strategy for relay selection, thereby enhancing overall communication quality while ensuring fairness among users.

INDEX TERMS Cooperative communication, uncrewed aerial vehicle (UAV) networks, relay selection, outage probability, game theory.

I. INTRODUCTION

With the rapid evolution of mobile communication technology and the high-speed development of the intelligent manufacturing industry, the user demand for the use of wireless services has significantly increased, and the requirements for the quality of service have been gradually improved [1]. Next-generation wireless communication networks are expected to offer higher data rates than 5G, while facing many challenges such as massive device access, ultra-high frequency spectrum efficiency, low latency, and higher link reliability [2], [3]. While in the wireless transmission process due to the transmission medium and the propagation environment, the signal quality may deteriorate in the transmission process and fail to meet the user's needs. Therefore realizing reliable signal transmission is important

for the research and development of wireless communication systems.

Cooperative communication is an efficient way of communication, for users with poor communication conditions, cooperative communication technology through the deployment of relay nodes to participate in the formation of virtual MIMO (Multiple-Input Multiple-Output) antenna arrays to form a multi-input multi-output system can effectively enhance the channel capacity, to combat channel fading and improve the stability of the link [4]. However, the relay deployment location of traditional collaborative communication systems built based on fixed scenarios is generally unchanged, and a large number of relay removals will result in a waste of resources as well as an increase in maintenance costs, with a weaker ability to cope with mobile scenarios.

Driven by the demand for higher system capacity and more flexible mobility management, UAVs have become vital for realizing ubiquitous connectivity in next-generation networks [5], [6]. Their role has evolved from initial aerial

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photography to providing auxiliary communications within cellular networks, establishing them as a major wireless research hotspot. This is evident in emerging applications such as UAV-assisted mobile edge computing and UAV-mounted reconfigurable intelligent surfaces for improving signal coverage and quality [7]. Due to its high mobility and wide coverage after networking, it is often used as an airborne relay to relay information from ground users [8]. For scenarios such as mountainous forests where ground relays are difficult to deploy, the flexibility of UAVs can break the limitations of traditional fixed relay communication ranges. The application of UAVs as airborne relays in collaborative communication technology can effectively reduce the probability of outage in communication links and improve system reliability and stability [9]. UAV service ground users are usually in the form of networks rather than separate individuals, with higher robustness of UAV service networks. In the case that the UAV serving the user is unable to continue to provide communication connectivity due to a failure, the user can switch to other nodes in the network to continue to provide communication services [10], [11]. Consequently, the research on cooperative communication networks with UAVs is of great application importance for complex communication environments. Among them, the selection of relay UAVs and service users directly affects the performance of the cooperative communication network, while a reasonable relay selection method not only improves the overall system capacity to reflect the advantages of cooperative diversity, but also improves the differences in the communication conditions between the users, and carries out a reasonable allocation of communication resources to avoid unnecessary resources wasted by relay competition between users. Therefore, the method of selecting relay nodes in UAV cooperative communication networks has become a hot research topic among scholars in recent years.

II. RELATED WORK AND CONTRIBUTIONS

A. RELATED WORK

In 2003, Sendonaris first proposed a communication model for three nodes in cooperative communication [12]. By now, the relay node selection method has become a hot issue in cooperative communication networks, and scholars have been working to find a reasonable method to balance the node selection of each user. Relay selection focuses on selecting suitable communication nodes to increase the system capacity, thereby reducing the probability of outage of the communication link and thus maintaining the stability of the communication.

Aljadai et al. proposed an optimal collaborative relay selection method under two-way two-hop cooperative communication networks and derived a closed expression for the outage probability based on the probability density function [13]. The method can achieve the approximate minimum outage probability under the proposed bidirectional two-hop cooperative communication network. In [14], the authors consider a cooperative NOMA (Non-Orthogonal

Multiple Access) system under cognitive radio and propose a maximum-minimum relay selection scheme applicable to this system, and test the performance of the proposed algorithm in terms of BER (Bit Error Rate) and outage probability through simulation experiments. In [15], the authors address the low-cost fixed clustering problem in wireless sensor networks by assigning different random relay vectors to each ring region in the network. All the relay vectors are combined into a random relay selection matrix, which randomly forwards data to the relay nodes through the probability distribution in the random relay selection matrix, thus reducing the deployment cost of wireless sensor networks. In [16], authors propose a deep Q-network-based relay selection scheme for use under wireless sensor networks, modeling cooperative communication in wireless sensor networks as a Markov decision process, the best relay is adaptively selected among the set of candidate relays when the instantaneous channel state information is unknown. In [17], the authors propose a region-aware relay selection strategy to study the reliability problem of relay networks, derive a closed-form expression for the reliability-enhancing region, and conduct simulation experiments to verify the reliability of the proposed method considering the randomness of the relay distribution. Reference [18] proposes an adaptive relay node selection method based on exponential segmentation based on the spatial location information of nodes to achieve message broadcasting for cooperative communication in complex scenarios. Reference [19] proposed parallel joint optimal relay selection algorithm based on instantaneous channel state information for wireless sensor networks with wake-up radio to reduce delay and energy consumption. In [20], a risk-aware relay selection scheme using reinforcement learning is proposed to ensure sustainable data transmission and vehicle safety in vehicular cooperative communication networks. An improved Kuhn-Munkres algorithm based on risk assessment is used to achieve relay selection in multi-relay and multi-target vehicle scenarios. Reference [21] proposes a max-min relay selection scheme for collaborative NOMA systems and evaluates the outage performance of the system.

Most of the above methods have been studied for traditional terrestrial cooperative communication networks. With the increasing popularity of civil UAVs, the problem of relay selection under UAV networks has gradually become an emerging topical issue.

Reference [22] addresses the accessibility of mobile devices under UAV-assisted communication networks by proposing a cooperative game approach for UAV relay selection during the switching process. Reference [23] addresses the problem of minimum outage probability for cellular users in UAV-assisted cooperative communication networks by proposing a relay selection method based on the optimal harmonic mean (HM), and demonstrates through simulation experiments that the algorithm has a better outage performance with different numbers of UAVs. Reference [24] aiming at the problem of limited

resources of UAV networks, a selection method based on distance and node correlation is proposed to balance the selection rate and reduce the collision probability. In [25], the authors address the problem of minimum energy consumption in uncrewed trunked communication networks by proposing a relay selection strategy that reduces the complexity of the algorithm and reduces the energy consumption of the network by calculating the angle between the UAV and the user with multiple fitness preselection and power constraints on the outage probability.

Most of the existing studies focus on maximizing system capacity or minimizing energy consumption and minimizing outage probability. However, little attention has been paid to the fairness problem of channel capacity increase brought about by different users after the selected relay and the interference problem brought about by the communication link during the relay selection process. Though these aspects can assess the performance of the network as a whole, it is equally important to avoid excessive differences in the communication conditions of different users. In addition, communication models that take into account interference in communication links are more relevant.

B. CONTRIBUTIONS

Based on the above analysis, this paper deploys UAVs as airborne relays in the cooperative communication network, and completes the collaborative communication between UAVs and ground users through the proposed relay selection method, to achieve the improvement of system throughput and guarantee the fairness of users. The following is the specific work:

- (1) To provide UAV relay services to ground users, a UAV-assisted cooperative communication network with multiple UAVs and multiple ground user pairs was first established. To make the proposed network more realistic, consider interference between direct and relay links for different users. The transmitted signal is divided by power into two parts, one for information such as the user's positional environment that is uploaded to the UAV, and one for information that needs to be sent to the destination user.
- (2) The integral expression for the outage probability under this system is derived from the derivation of the system model, as the system model differs from the traditional cooperative communication model due to the consideration of the co-channel interference between the direct transmission links as well as the relay links and the new signal form.
- (3) Under the simultaneous consideration of maximizing system capacity and fairness among users, the relay selection process is modeled as a network formation game model, and a decentralized search relay selection algorithm considering system capacity and fairness among users is proposed to solve the game model

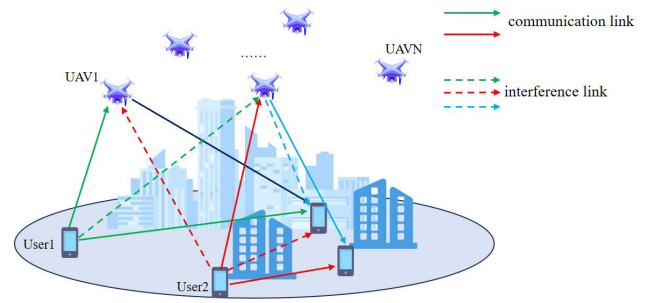


FIGURE 1. UAV Air-to-Ground cooperative communication networks model.

to obtain an equilibrium relay selection strategy. The initial policy set is first generated by the diverse relay policy set generation method. Secondly, the selection process is optimized by alternating iterations through a range of iterations, and conflicts between users are avoided by a neighborhood machine search strategy. Finally, the proposed method is verified by simulation experiments in terms of system capacity, probability of outage for each user, and user fairness and compared with the remaining five methods.

III. SYSTEM MODELLING AND PERFORMANCE ANALYSIS

In the UAV relay cooperative communication network, it is assumed that the relay collaboration process increases the channel capacity and the relay forwarding is in the form of DF (Decode and Forward). It is assumed that the UAV collects state information (position state information, channel state information, etc.) from the ground user at regular intervals. Assuming that the user needs to upload its status information each time a relay UAV is employed for forwarding, the UAV is considered to be hovering for the time slot of information forwarding. The co-channel interference generated from other nodes during message forwarding is considered and the signal-to-interference-plus-noise ratio is used to measure the interference suffered during the communication process. Each UAV serves only one pair of ground users at the same moment and adopts half-duplex transceiver mode. For the purpose of this analysis, it is considered that users use the relay UAV service in the form of user pairs, the i -th source user communicates only with the i -th destination user. The system model schematic is shown in Fig. 1.

A. CHANNEL MODEL

In complex urban environments, the Line-of-Sight (LoS) channel between ground users and UAVs is prone to obstruction by tall buildings. Therefore, this study employs a channel model that incorporates both small and large-scale fading, effectively representing scenarios where the LoS link is either unavailable or negligible due to the UAV's low flight altitude [26].

$$h_{s_i u_j} = \frac{g_{s_i u_j}}{\sqrt{1 + d_{s_i u_j}^\alpha}} \quad (1)$$

$$h_{u_i d_j} = \frac{g_{u_i d_j}}{\sqrt{1 + d_{u_i d_j}^e}} \quad (2)$$

$$h_{s_i d_i} = \frac{g_{s_i d_i}}{\sqrt{1 + d_{s_i d_i}^e}} \quad (3)$$

The channel gain from source user i to UAV j is given by $h_{s_i u_j}$, where $g_{s_i u_j}$ is the small-scale fading coefficient and $\frac{1}{\sqrt{1 + d_{s_i u_j}^e}}$ models the large-scale path loss. Similarly, the channel gain from UAV i to destination user j is $h_{u_i d_j}$, and the direct link gain from source user i to destination user i is $h_{s_i d_i}$. e is the power exponential path loss factor.

B. TRANSMISSION MODEL

In the first phase, the source user forwards the signals to the destination user and the UAV relay respectively. Where the i -th source broadcasts a synthesized signal:

$$x_i = \sqrt{a_1 P_s} x_{u_i} + \sqrt{a_2 P_s} x_{d_i} \quad (4)$$

where x_{u_i} is the part of the signal uploaded by the source user to the UAV relay network and x_{d_i} is the part of the signal sent by the source user to the destination user. P_s is the transmission power, assuming that all source user nodes have the same transmission power. a_1, a_2 is the power control factor.

The signal transmitted from source user i to destination user i is:

$$y_{s_i d_i} = h_{s_i d_i} [\sqrt{a_1 P_s} x_{u_i} + \sqrt{a_2 P_s} x_{d_i}] + n_{s_i d_i} \quad (5)$$

The signal transmitted from source user i to UAV j is:

$$y_{s_i u_j} = h_{s_i u_j} [\sqrt{a_1 P_s} x_{u_i} + \sqrt{a_2 P_s} x_{d_i}] + n_{s_i u_j} \quad (6)$$

Assuming that relay UAV can fully extract the signal from the source c forwarded to the destination user j by Successive Interference Cancellation(SIC) technique is

$$y_{u_i d_j} = \sqrt{P_{u_i}} h_{u_i d_j} \sqrt{a_2 P_s} x_{d_i} + n_{u_i d_j} \quad (7)$$

where $n_{s_i d_i}$, $n_{s_i u_j}$, $n_{u_i d_j}$, are additive Gaussian white noises with mean 0 and variance σ^2 . P_{u_i} is the transmit power of the relay UAV i .

C. CHANNEL CAPACITY ANALYSIS

During direct transmission, the transmission link Signal-to-Interference-plus-Noise Ratio (SINR) from source user i to destination user i is:

$$\text{SINR}_{s_i d_i} = \frac{\frac{g_{s_i d_i}^2}{1 + d_{s_i d_i}^e} a_2 P_s}{\frac{g_{s_i d_i}^2}{1 + d_{s_i d_i}^e} a_1 P_s + \sum_{s_k \in F_{s_i d_i}, k \neq i} P_s \frac{g_{s_k d_i}^2}{1 + d_{s_k d_i}^e} + \sigma^2} \quad (8)$$

where $F_{s_i d_i}$ is the set of co-channel interferences that the destination user i is subjected to by the source user. $s_k \in F_{s_i d_i}$ denotes a source user belonging to the set $F_{s_i d_i}$. $\sum_{s_k \in F_{s_i d_i}, k \neq i} P_s \frac{g_{s_k d_i}^2}{1 + d_{s_k d_i}^e} + \sigma^2$ is the co-channel interference

caused by other source users during direct transmission from the source user i to the destination user i [27].

The transmission link Signal-to-Interference-plus-Noise Ratio (SINR) from source user i to UAV j is:

$$\text{SINR}_{s_i u_j} = \frac{\frac{g_{s_i u_j}^2}{1 + d_{s_i u_j}^e} P_s}{\sum_{s_k \in F_{s_i u_j}, k \neq i} P_s \frac{g_{s_k u_j}^2}{1 + d_{s_k u_j}^e} + \sigma^2} \quad (9)$$

where $F_{s_i u_j}$ is the set of co-channel interferences that the relay UAV j is subjected to by the source user. $s_k \in F_{s_i u_j}$ denotes a source user belonging to the set $F_{s_i u_j}$. $\sum_{s_k \in F_{s_i u_j}, k \neq i} P_s \frac{g_{s_k u_j}^2}{1 + d_{s_k u_j}^e}$ is the co-channel interference caused by other source users to relay UAV j while the source user i is transmitting to it.

In the relay forwarding phase, the transmission link SINR from relay UAV i to destination user j is:

$$\text{SINR}_{u_i d_j} = \frac{\frac{g_{u_i d_j}^2}{1 + d_{u_i d_j}^e} P_s a_2 P_{u_i}}{\sum_{u_k \in F_{u_i d_j}, k \neq i} P_{u_i} \frac{g_{u_k d_j}^2}{1 + d_{u_k d_j}^e} + \sigma^2} \quad (10)$$

where $F_{u_i d_j}$ is the set of co-frequency interferences of the destination user j by the relay UAV. $u_k \in F_{u_i d_j}$ denotes a relay UAV user belonging to the set $F_{u_i d_j}$. $\sum_{u_k \in F_{u_i d_j}, k \neq i} P_{u_i} \frac{g_{u_k d_j}^2}{1 + d_{u_k d_j}^e}$ is co-channel interference caused by other UAV retransmissions while relay UAV i is decoding and retransmitting to destination user j .

Therefore, the channel capacity from source user i to destination user i is:

$$C_{s_i d_i} = B_W \log_2 (1 + \text{SINR}_{s_i d_i}) \quad (11)$$

The channel capacity from source user i to UAV j is:

$$C_{s_i u_j} = B_W \log_2 (1 + \text{SINR}_{s_i u_j}) \quad (12)$$

The channel capacity from UAV i to destination user j is:

$$C_{u_i d_j} = B_W \log_2 (1 + \text{SINR}_{u_i d_j}) \quad (13)$$

Since SIC is performed at the relay UAV, which can be considered as a decode-and-forward(DF) system, the capacity of the cooperative communication system for the i -th pair of users via relay UAV j is:

$$C_{s_i} = \frac{1}{2} \min \left\{ W \log_2 (1 + \text{SINR}_{s_i u_j}), W \log_2 (1 + \text{SINR}_{s_i d_i} + \text{SINR}_{u_i d_j}) \right\} \quad (14)$$

where B_W represents the bandwidth.

D. OUTAGE ANALYSIS

A cooperative communication system under the DF forwarding protocol is considered to have a communication link outage when both the direct and relay transmission links are outage at the same time. Consider the interruption of a particular user, define A event occurs as a communication link

outage between the source and the destination, event B as a communication link outage between the source and the relay UAV, and event C as a communication link outage between the relay UAV and the destination. Then the outage event of the system can be represented as:

$$P_{out}^{DF} = P(A) (1 - P(\overline{BC})) \quad (15)$$

where $P(\overline{BC})$ represents that no outage occurred in both the receive and forward links of the relay UAV, the relay communication was completed. The event B is independent of the event C . The outage event can be represented as:

$$P_{out}^{DF} = P(A) (1 - P(\overline{BC})) \\ = (1 - P(\overline{A})) (1 - P(\overline{B}) P(\overline{C})) \quad (16)$$

For ease of analysis, let the bandwidth $B_W = 1$. The probability of successful transmission of the three links is:

$$P(\overline{A}) = P(SINR_{s_i d_i} \geq 2^{2R_{th}} - 1) \quad (17)$$

$$P(\overline{B}) = P(SINR_{s_i u_j} \geq 2^{2R_{th}} - 1) \quad (18)$$

$$P(\overline{C}) = P(SINR_{u_j d_j} \geq 2^{2R_{th}} - 1) \quad (19)$$

Let $\gamma_2 = \sum_{s_k \in F_{s d_i}, k \neq i} P_s \frac{g_{s_k d_i}^2}{(1 + d_{s_k d_i}^e) \sigma^2}$, $\gamma_1 = \frac{g_{s_i d_i}^2}{(1 + d_{s_i d_i}^e) \sigma^2} P_s$, Eq.(8) can be expressed in the form of a random variable combined with random variables γ_1, γ_2 as shown in Eq.(20).

$$SINR_{s_i d_i} = \frac{a_2 \gamma_1}{a_1 \gamma_1 + \gamma_2 + 1} \quad (20)$$

Assuming that all Rayleigh fading channels are independent of each other, and suppose that the PDF of $g_{s_k d_i}^2$ is $\frac{1}{\mu_{s_i d_i}} \exp(-\frac{x}{\mu_{s_i d_i}})$, By Theorem 2.1, the probability density function of γ_2 is a linear combination of n exponential distribution functions. The PDF of γ_2 is given by $f(x) = \sum_{j=1}^n c_j \rho_j + c_1 \rho_1 + c_2 \rho_2 + \dots + c_n \rho_n$, $c_k = \frac{\prod_{j \neq k} \lambda_j}{\prod_{j \neq k} (\lambda_j - \lambda_k)}$, $\rho_k(x) = \lambda_k e^{-\lambda_k x}$,

$$\lambda_k = \frac{\mu_{s d} \sigma^2 + d_{s_k d_i}^e \mu_{s d} \sigma^2}{g_{s_k d_i}^2 P_s}.$$

Theorem 1: The probability density of a new random variable consisting of the sum of n random variables that are independent and obey different exponential distributions is a linear combination of the density functions of these n exponential distributions (see the appendix for the proof).

Let $Z = \frac{a_2 \gamma_1}{a_1 \gamma_1 + \gamma_2 + 1}$, $W = a_1 \gamma_1 + \gamma_2 + 1$. γ_1 and γ_2 are independent of each other and clearly have a unique inverse function. By applying the change of variables technique to the double integral, the Jacobian determinant of the transformation is given by Eq.(21)

$$J = \begin{vmatrix} \frac{\partial \gamma_1}{\partial z} & \frac{\partial \gamma_2}{\partial z} \\ \frac{\partial \gamma_1}{\partial w} & \frac{\partial \gamma_2}{\partial w} \end{vmatrix} = \begin{vmatrix} \frac{w}{a_2} & -\frac{a_1}{a_2} w \\ 1 & 1 - \frac{a_1}{a_2} z \end{vmatrix} = w + \left(\frac{a_1}{a_2^2} - \frac{a_1}{a_2} \right) w z \quad (21)$$

Then the joint probability density function of Z given by Eq.(22).

$$\rho(z, w) = \rho(\gamma_1(z, w), \gamma_2(z, w)) |J| \quad (22)$$

Since γ_1 and γ_2 are independent of each other, the joint probability density function of γ_1 and γ_2 is the product of their respective probability density functions. Substituting the substituted variables into Eq.(22) yields Eq.(23).

$$\rho(z, w) \\ = \lambda_0 w e^{-\lambda_0 \frac{1}{a_2} z w} \sum_{k=1}^{\infty} c_k \lambda_k e^{-\lambda_k \left(w - \frac{a_1}{a_2} z w - 1 \right)} \\ + \left(\frac{a_1}{a_2^2} - \frac{a_1}{a_2} \right) w z \lambda_0 e^{-\lambda_0 \frac{1}{a_2} z w} \sum_{k=1}^{\infty} c_k \lambda_k e^{-\lambda_k \left(w - \frac{a_1}{a_2} z w - 1 \right)}. \quad (23)$$

where λ_0 stands for the parameter of the exponential distribution obeyed by the square of the channel gain of the computed user pair.

To derive the marginal probability density of z , we integrate the expression $\int_{-\infty}^{+\infty} \rho(z, w) dw$. Because of $W = a_1 \gamma_1 + \gamma_2 + 1$, γ_1, γ_2 are non-negative random variables, $a_1 \geq 0$. It can be concluded that $W \geq 1$. Adjusting the lower limit of integration to 1 and collapsing gives:

$$\int_{-\infty}^{+\infty} \rho(z, w) dw \\ = \int_1^{+\infty} \rho(z, w) dw \\ = \sum_{k=1}^{\infty} \frac{a_2 \lambda_0 c_k \lambda_k e^{\lambda_k}}{\lambda_0 z - \lambda_k a_1 z + \lambda_k a_2} \left(1 + \frac{a_2}{\lambda_0 z - \lambda_k a_1 z + \lambda_k a_2} \right) \\ \times e^{-\left(\frac{\lambda_0 - \lambda_k a_1}{a_2} z + \lambda_k \right)} \left(1 + \left(\frac{a_1}{a_2^2} - \frac{a_1}{a_2} \right) z \right) \quad (24)$$

Let $\gamma_3 = \frac{g_{s_i u_j}^2}{1 + d_{s_i u_j}^e} P_s$, $\gamma_4 = \sum_{s_k \in F_{s u_j}, k \neq i} P_s \frac{g_{s_k u_j}^2}{(1 + d_{s_k u_j}^e) \sigma^2}$, Eq.(9) can be expressed as a random variable in the form of a combination of random variables γ_3, γ_4 as shown in Eq.(25).

$$SINR_{s_i u_j} = \frac{\gamma_3}{\gamma_4 + 1} \quad (25)$$

γ_3, γ_4 are the same type of random variables as γ_1, γ_2 , respectively, they have the same form of probability density. $\gamma_3 \sim \exp\left(\frac{\mu_{s u} \sigma^2 + d_{s_k u_j}^e \mu_{s u} \sigma^2}{g_{s_k u_j}^2 P_s}\right)$, The PDF of γ_4 is given

by $f(x) = b_1 \tau_1 + b_2 \tau_2 + \dots + b_n \tau_n$, $b_k = \frac{\prod_{j \neq k} \eta_j}{\prod_{j \neq k} (\eta_j - \eta_k)}$,

$$\tau_k(x) = b_k e^{-b_k x}, \eta_k = \frac{\mu_{s u} \sigma^2 + d_{s_k u_j}^e \mu_{s u} \sigma^2}{g_{s_k u_j}^2 P_s}.$$

Let $Z = \frac{\gamma_3}{\gamma_4 + 1}$, $W = \gamma_4 + 1$. The transformed Jacobi determinant is obtained as Eq.(26) by the method of transforming variables of the dual integral.

$$J = \begin{vmatrix} \frac{\partial \gamma_3}{\partial z} & \frac{\partial \gamma_4}{\partial z} \\ \frac{\partial \gamma_3}{\partial w} & \frac{\partial \gamma_4}{\partial w} \end{vmatrix} = \begin{vmatrix} w & 0 \\ z & 1 \end{vmatrix} = w \quad (26)$$

The transformed joint probability density is consistent with the form of Eq.(22), so γ_3 and γ_3 are mutually independent random variables whose joint probability densities are the product of their respective probability densities.

$$\rho(z, w) = \eta_0 e^{-\eta_0 z w} \sum_{k=1} b_k \eta_k e^{-\eta_k (w-1) w} \quad (27)$$

where η_0 is substituted for a parameter of the exponential distribution obeyed by the square of the channel gain of the selected relay to the source.

To derive the marginal probability density of z , we integrate the expression $\int_{-\infty}^{+\infty} \rho(z, w) dw$. Due to $W = \gamma_4 + 1$, γ_3 , γ_4 are non-negative random variables, it can be concluded that $W \geq 1$. Adjusting the lower limit of integration to 1 and collapsing gives:

$$\begin{aligned} \int_{-\infty}^{+\infty} \rho(z, w) dw &= \int_1^{+\infty} \rho(z, w) dw \\ &= \sum_{k=1} (\eta_0 b_k \eta_k e^{\eta_k} - 1) \frac{e^{-(\eta_0 z + \eta_k)}}{\eta_0 z + \eta_k} \end{aligned} \quad (28)$$

Let $\gamma_5 = \frac{g_{u_i d_j}^2}{(1+d_{u_i d_j}^e) \sigma^2} P_s a_2 P_{u_i}$, $\gamma_6 = \sum_{u_k \in F_{udj}, k \neq i} P_{u_i} \frac{g_{u_k d_j}^2}{(1+d_{u_k d_j}^e) \sigma^2}$. Eq.(10) can be expressed as a random variable in the form of a combination of random variables γ_5 , γ_6 as shown in Eq.(29).

$$SINR_{u_i d_j} = \frac{\gamma_5}{\gamma_6 + 1} \quad (29)$$

where $\gamma_5 \sim \exp(\frac{\mu_{ud} \sigma^2 + d_{s_k u_j}^e \mu_{ud} \sigma^2}{a_2 g_{s_k u_j}^2 P_s P_{u_i}})$, the PDF of γ_6 is given by

$$f(x) = a_1 \varepsilon_1 + a_2 \varepsilon_2 + \dots + a_n \varepsilon_n, a_k = \frac{\prod_{j \neq k} \theta_j}{\prod_{j \neq k} (\theta_j - \theta_k)}, \theta_k = \frac{\mu_{ud} \sigma^2 + d_{s_k u_j}^e \mu_{ud} \sigma^2}{a_2 g_{s_k u_j}^2 P_s P_{u_i}}.$$

Eq.(29) has the same structure as Eq.(25) and the same reasoning leads to:

$$\begin{aligned} \int_{-\infty}^{+\infty} \rho(z, w) dw &= \int_1^{+\infty} \rho(z, w) dw \\ &= \sum_{k=1} (\theta_0 a_k \theta_k e^{\theta_k} - 1) \frac{e^{-(\theta_0 z + \theta_k)}}{\theta_0 z + \theta_k} \end{aligned} \quad (30)$$

The outage probability of the system is derived from equations (15), (17), (18), (19), (24), (28), and (30):

$$\begin{aligned} P_{out}^{DF} &= \int_0^{r_{th}} \sum_{k=1} \frac{a_2 \lambda_0 c_k \lambda_k e^{\lambda_k}}{\lambda_0 z - \lambda_k a_1 z + \lambda_k a_2} e^{-\left(\frac{\lambda_0 - \lambda_k a_1}{a_2} z + \lambda_k\right)} \\ &\times \left(1 + \frac{a_2}{\lambda_0 z - \lambda_k a_1 z + \lambda_k a_2}\right) \left(1 + \left(\frac{a_1}{a_2^2} - \frac{a_1}{a_2}\right) z\right) \\ &\times \left(1 - \left(1 - \sum_{k=1} (\eta_0 b_k \eta_k e^{\eta_k} - 1) \frac{e^{-(\eta_0 z + \eta_k)}}{\eta_0 z + \eta_k}\right)\right) \\ &\times \left(\left(1 - \sum_{k=1} (\theta_0 a_k \theta_k e^{\theta_k} - 1) \frac{e^{-(\theta_0 z + \theta_k)}}{\theta_0 z + \theta_k}\right)\right) dz \end{aligned} \quad (31)$$

This expression elucidates the key parameters governing the outage probability. Nevertheless, owing to the intractable nature of its integral form, the subsequent performance analysis is primarily conducted using Monte Carlo simulations.

E. USER FAIRNESS ANALYSIS

In statistical analysis, variance is used to calculate the difference between each variable and the overall mean, and is a measure of the degree of dispersion of a random variable or set of data. In cooperative communication systems, the channel capacity of the user determines the quality of the communication. Therefore, the fairness of each user in the relay selection process is measured in the form of variance.

The average channel capacity of users under the UAV cooperative communication network can be expressed as:

$$\bar{C}_n = \frac{\sum_{n=1}^N C_{s_n}}{N} \quad (32)$$

The variance of the channel capacity of all users concerning the average channel capacity is:

$$F_{var} = \frac{1}{N} \sum_{n=1}^N (C_{s_n} - \bar{C}_n)^2 \quad (33)$$

Eq. (33) is the measure of user fairness in the relay selection process under the UAV cooperative communication network.

IV. GAME MODEL AND EQUILIBRIUM ANALYSIS

A. GAME MODEL

Through the above system model and performance analysis, the UAV network relay selection problem to be solved is established as a network formation game model. The network formation game focuses on the process in which multiple players are influenced by the network topology to form a communication network by selecting communication nodes through a game. The nodes are in a non-cooperative relationship with each other to ensure their respective communication quality. To solve the UAV relay node selection problem using a network formation game, it is necessary to first model the problem as a network formation game [28].

The game model is represented as $G:V$ denotes the set of participants; $G = \{V, \{S_i\}_{i \in V}, \{u_i\}_{i \in V}\}$, S_i denotes the set of actions that the participant i can choose; u_i denotes the benefits that the participants get after choosing the actions, and different actions will bring different benefits. The meanings of the components represented in the relay selection problem are as follows:

1) PARTICIPANT SET V

The N pairs of users in the UAV cooperative communication network are taken as participants in the game, so the set of participants in the network formation game can be represented as:

$$V = \{1, 2, \dots, N\} \quad (34)$$

2) ACTION SET S_i

In the relay selection problem, the UAV selected by the user requiring cooperative communication is used as the action of the user pair. $S_i(G) = \{(i, j)\}$ represents that the relay UAV selected by the i -th user is the j -th one. To avoid preemptive relaying, the actions of different participants in the same decision-making round cannot be the same. Each user pair in this game model chooses a relay UAV as their only action for each decision round. This is a simultaneous, static decision-making process. As a result, the entire collection of possible UAVs defines both the strategy space and the action space for every given user pair.

3) UTILITY U_i

After the user selects the UAV node for cooperative communication, the gain of the user pair is set to be the channel capacity after collaborative forwarding by the UAV based on the performance metrics analyzed above:

$$u_i = C_{s_i} \quad (35)$$

Using two UAVs and two user pairs as an example, the utility matrix is given by Eq.(36).

$$U_{2users2uavs} = \begin{bmatrix} (0, 0) & (C_{s_1}^{u_1}, C_{s_2}^{u_2}) \\ (C_{s_1}^{u_2}, C_{s_2}^{u_1}) & (0, 0) \end{bmatrix} \quad (36)$$

The matrix elements are defined as follows: $(C_{s_1}^{u_2}, C_{s_2}^{u_1})$ denotes the utility when user 1 selects UAV 2 and user 2 selects UAV 1, while $(C_{s_1}^{u_1}, C_{s_2}^{u_2})$ denotes the utility for the reverse assignment. Element $(0, 0)$ represents a utility of zero, which occurs when both users select the same relay UAV, violating the system constraint.

B. NASH EQUILIBRIUM ANALYSIS OF GAMES

The network formation game model for relay selection is a non-cooperative game model [29]. The participants are N pairs of finite users and the set of actions S_i is M drones capable of providing relay services. Based on the above conditions, the game is an N -player non-cooperative finite game. By Theorem 3.1, the game must have a Nash equilibrium.

Theorem 2: Nash equilibria must exist for n -player non-cooperative finite games [30].

V. PROBLEM FORMULATION

The network formation game is a non-cooperative game, each participant is committed to achieving their maximum gain, to a certain extent, to ensure the network fairness. Therefore, the maximisation of the overall network revenue is the optimisation goal, to establish the following objective function:

$$P : S^* = \arg \max \sum_{i=1}^M u_i \quad (37)$$

where S^* represents the set consisting of the relay indexes selected by each user with the following constraints:

$$s.t. C1 : S^* \subseteq N \quad (38)$$

$$C2 : |S^*| = M \quad (39)$$

Eq.(38) indicates that the sought relay set must be included in the UAV set N , i.e., the selected UAV serial number must exist. Eq.(39) indicates that the length of the desired set of relays must be M and M is the number of user pairs. That is, each pair of users has a corresponding UAV relaying them. In addition, the mutual dissimilarity of the elements in the set ensures that a UAV serves only one pair of users, and there is no problem of competition between users for UAV relaying.

VI. ALGORITHM DESIGN

In order to solve the proposed game problem of relay UAV selection among users, based on the idea of systematic construction of new solutions and deconvolution in scatter search algorithms to improve concentration and diversity in the search process [31], we designed a scatter search relay selection algorithm that considers the system capacity as well as user fairness(SSRS). The algorithm generates the initial strategy matrix and updates it through the diversified strategy method, and optimises the system gain and user fairness alternately according to the number of iterations, so as to achieve the purpose of maximising the system gain and the fairness of the gain among the users, the specific steps are as follows.

Step1, Initialise the system parameters and enter the number of user pairs M , the number of relay UAVs N , the power allocation factors a_1, a_2 , and the maximum number of iterations of the algorithm $iter_{max}$.

Step2, Diversified generation of initial relay selection strategy set. According to the proposed diversified relay strategy set generation method, the initial relay selection strategy set is generated.

Step3, Calculate the system gains corresponding to each strategy in the set of relay selection strategies as well as user fairness, and determine the elite strategies retained based on the range of iteration counts.

Step4, The retained policies will be used to generate a new set of policies according to the diverse relay strategy set generation method.

Step5, The optimal relay selection policy is output when the maximum number of iterations is reached.

The pseudo-code design of the algorithm is as follows.

In which to increase the diversity of initial relay selection strategies as well as to improve the retained elite strategies, we propose a diverse relay strategy set generation method. The method replaces some user-selected relay UAV nodes by swapping, random replacement, inversion, and neighbourhood search replacement methods for the policies in the input policy matrix, and generates a new policy set by mixing the sub-policy sets generated by the replacement in a certain ratio. The pseudo-code design of the algorithm is as follows.

In order to avoid the phenomenon of relay seizure in the relay selection strategy generated during the iterative process of the algorithm, a neighbourhood UAV search replacement method is proposed, which can effectively solve the relay competition problem and increase the diversity of strategies

Algorithm 1 A Scatter Search Relay Selection Algorithm That Considers the System Capacity as Well as User Fairness

Input: Number of user pairs M , Number of UAVs N , power allocation factors a_1, a_2 , Maximum Iterations $iter_{max}$

Output: Relay strategy s

- 1: Initialization UAV position set D_u , User's location set D_s , channel information matrix H_{su}, H_{ud}, H_{sd} ;
 - 2: Randomly initialise the relay strategy matrix and build the initial relay strategy matrix S according to the diversified relay strategy set generation approach;
 - 3: **repeat**
 - 4: **for** each row of strategies in the strategy matrix S **do**
 - 5: Calculate the benefit function u , the system variance fairness F_{var} , the total system benefit U for each user under this strategy from Eq.(33), Eq.(35);
 - 6: **end for**
 - 7: Sort U in descending order, Sort F_{var} in ascending order;
 - 8: **if** the goal of selecting a relay for the current range of iterations is the system gain **then**
 - 9: Strategies for choosing reservations from U ;
 - 10: **else if** The goal of selecting a relay for the current range of iterations is system fairness **then**
 - 11: Strategies for choosing reservations from F_{var} ;
 - 12: **end if**
 - 13: Diversity generation of retained strategies to form a new strategy matrix S_{new} ;
 - 14: $S = S_{new}$;
 - 15: **until** $iter = iter_{max}$;
-

Algorithm 2 A Diverse Relay Strategy Set Generation Method

Input: S , Number of strategies generated by diversification m

Output: S_{new} ;

- 1: **for** each strategy in input S **do**
 - 2: Interchange some of these user-selected relay UAV nodes;
 - 3: Random replacement of some of these user-selected relay UAV nodes;
 - 4: Replacement of some user-selected relay UAV nodes according to the neighbourhood UAV search replacement method;
 - 5: Reverse some of these user-selected relay UAV nodes;
 - 6: **end for**
 - 7: Mix the newly generated strategies in a certain ratio to get the new strategy matrix S_{new} ;
 - 8: **for** each strategy in S_{new} **do**
 - 9: Applying neighbourhood UAV search replacement method to Repeatedly Selected UAV nodes;
 - 10: **end for**
-

by searching for idle UAVs near the seized relay UAV and assigning them to the seizing user.

The proposed algorithm can be decomposed into four primary stages: strategy evaluation, ranking, strategy selection,

Algorithm 3 A Neighbourhood UAV Search Replacement Method

- 1: **for** repeatedly selected UAV nodes or nodes that need to be replaced **do**
 - 2: Based on the number of preempted users minus one, the distance is prioritised to find other idle drones in the vicinity of that drone; verse some of these user-selected relay UAV nodes;
 - 3: **for** newly searched idle UAVs **do**
 - 4: Calculating the distance from an idle UAV to a grabbing user;
 - 5: Allocation of free UAVs to preemptive users based on the distance priority principle;
 - 6: **end for**
 - 7: **end for**
-

and strategy generation. The time complexity per iteration for each stage is as follows: strategy evaluation is $O(S \times N)$, ranking is $O(S \times \log S)$, strategy selection is $O(S)$, and strategy generation is $O(K \times neighbor \times M + S \times N)$. Where $neighbor$ represents the number of policies replaced during neighbor machine search. K denotes the number of input policies used for generating diversified policies. S represents the number of strategies in the strategy matrix. After combining like terms and neglecting lower-order terms, the time complexity for a single iteration simplifies to $O(S \times N) + O(S \times \log S) + O(S \times M)$.

The overall time complexity of the algorithm is $O(T \times (S \times N + S \times \log S + S \times M))$. The algorithm exhibits polynomial time complexity, governed by the size of the policy matrix and the UAV network.

VII. SIMULATION RESULTS AND ANALYSIS

A. SIMULATION PARAMETER

Simulation experiments of the proposed algorithm are conducted using MATLAB, considering that the users are randomly distributed in a horizontal plane range of 1000 metres 1000 metres, and the UAV provides relay services to the ground users in an altitude range from 15 metres to 150 metres, and the ground users complete the selection of relay UAVs in a short period, and it is considered that the UAVs are in a stationary hovering state during the selection process. Table 1 presents the simulation parameters.

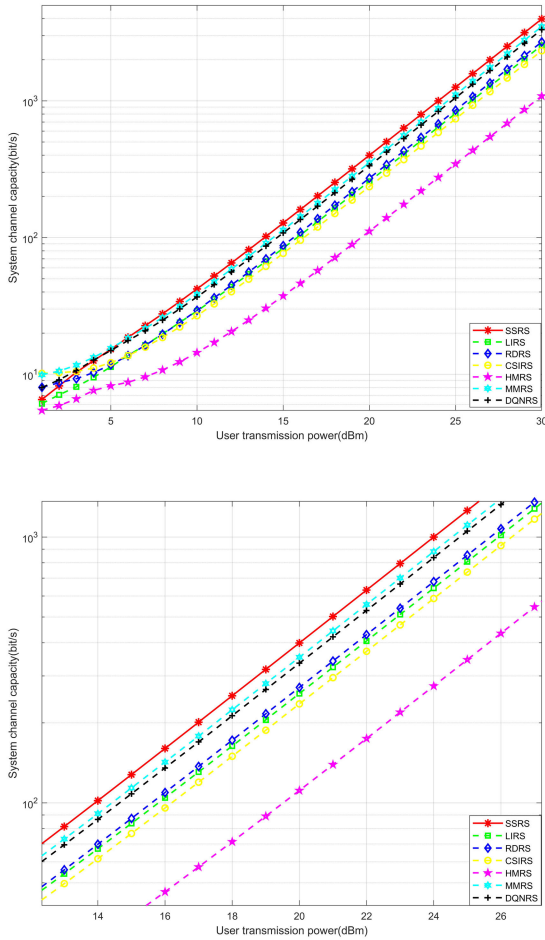
B. SIMULATION RESULTS AND ANALYSIS

The system capacity, the probability of outage for each user and the fairness of the normalised total system capacity variance are used as the main evaluation metrics. Selection of the current mainstream relay selection methods and the classical relay selection methods as comparison methods. Compare the proposed strategy with the following methods.

- (1) Random relay selection method (RDRS): This method randomly selects a UAV as a relay for cooperative communication from all the UAVs capable of providing

TABLE 1. Simulation parameter setting.

Simulation parameters	Value
Plane area coverage	$x_{\min} = y_{\min} = 0m, x_{\max} = y_{\max} = 1000m$
UAV altitude range	$z_{\min} = 15m, z_{\max} = 150m$
Number of UAVs	N
UAV transmission power	$P_u = 30\text{dBm}$
Maximum transmission power for ground users	$P_s = 30\text{dBm}$
Number of ground user pairs	M
Maximum number of iterations of the algorithm	$iter_{\max} = 500$
Power allocation factor	$a_1 = 0.4, a_2 = 0.6$
Path loss factor	$e = 4$
Noise power	$N_0 = -100\text{dBm}$

**FIGURE 2.** Comparison of system capacity under different relay selection methods($M = 3$ $N = 20$).

the service, regardless of the merits of the channel conditions and the distance.

- (2) Relay selection method based on location information (LIRS): This method selects the closest UAV node to the user as a relay node by using the collected location information.
- (3) Relay selection method based on channel state information (CSIRS): This method selects the UAV with the

best communication link conditions as a relay node for cooperative communication based on the instantaneous channel state information.

- (4) Relay selection method based on harmonic mean (HMRS) [18]: This method integrates the channel quality from the user to the UAV and from the UAV to the destination user, and selects the relay UAV for communication by calculating the harmonic mean.
- (5) Max–min relay selection method (MMRS) [16]: This approach performs cooperative communication by selecting relay UAVs that maximise the worst link performance.
- (6) Relay selection method based on Deep Q-learning (DQNRS) [32]: This method selects relay nodes through deep Q network.

Fig. 2 shows the variation of total system channel capacity with user's transmit power for six algorithms. From the simulation results, it is clear that the total channel capacity of the system increases with the increase in the user's transmit power. The proposed SSRS algorithm gives the best results, MMRS algorithm comes next and the HMRS algorithm gives the worst results. The total system channel capacity of the proposed algorithm is higher than the other five algorithms at all transmit power levels.

A horizontal comparison of the curves in Fig. 2 shows that the SSRS algorithm requires the lowest transmit power when trying to achieve the same total system channel capacity, and the transmit power required by the user is lower than the other algorithms by 0.3 dBm, 0.8dBm, 1.6 dBm, 1.8 dBm, and 2.3 dBm, respectively, when the total channel capacity is 10^2 bit/s.

A longitudinal comparison of the curves in Fig. 2 shows that when the user's transmit power is greater than 20 dBm, the total system channel capacity of all algorithms is greater than 10^2 bit/s, in which the total channel capacity of SSRS algorithm is 3.99×10^2 bit/s, which is higher than the other algorithms of 46.5 bit/s, 64.1 bit/s, 127.7 bit/s, 141.7 bit/s, 163.8 bit/s, and 288 bit/s, respectively. Compared to the MMRS algorithm, the total system capacity is increased by 13%.

Fig. 3 compares the user fairness performance of various relay selection methods across different source node transmit

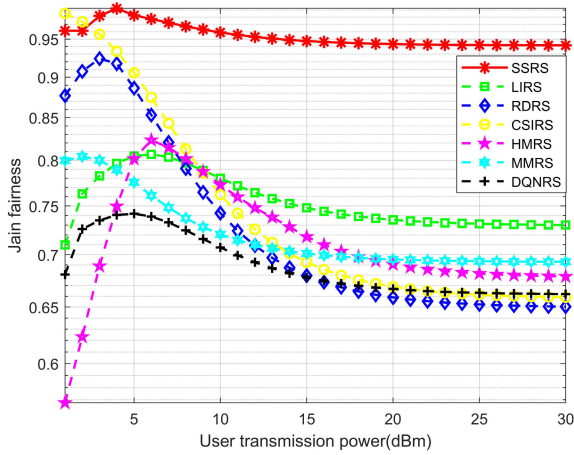


FIGURE 3. Comparison of system fairness under different relay selection methods ($M = 3, N = 20$).

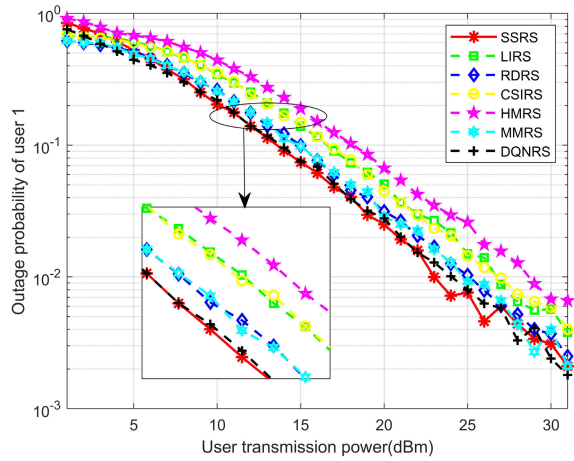


FIGURE 4. Comparison of outage probability of user 1 under different relay selection methods.

power levels. User fairness is quantified by the Jain Fairness Index based on channel capacity, as defined in Eq. (39). User fairness is quantified by the Jain's Fairness Index. A value closer to 1 at a given transmit power level indicates a fairer relay allocation, meaning that the communication performance among users in the network is more equitable. An index of 1 signifies absolute fairness, where each user achieves precisely the same communication rate through relay-assisted cooperation. Conversely, a value closer to 0 reflects a highly inequitable allocation of relay resources. This typically results from the selfish behavior of nodes, leading to significant disparities in communication conditions within the system.

$$F_{Jain} = \frac{\left(\sum_{i=1}^M C_{s_i} \right)^2}{M \sum_{i=1}^M (C_{s_i})^2} \quad (40)$$

As shown in Fig. 3, the fairness index of the proposed algorithm fluctuates when the transmit power

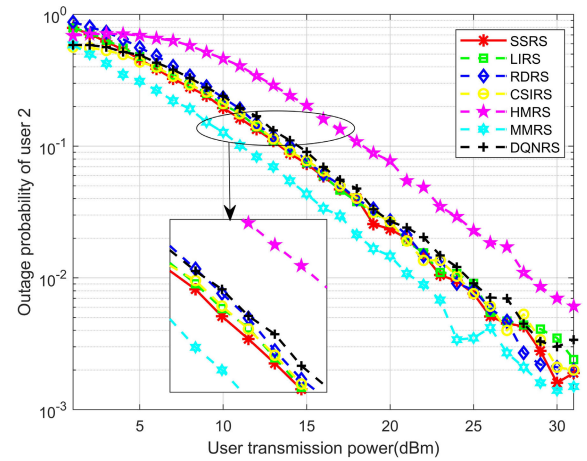


FIGURE 5. Comparison of outage probability of user 2 under different relay selection methods.

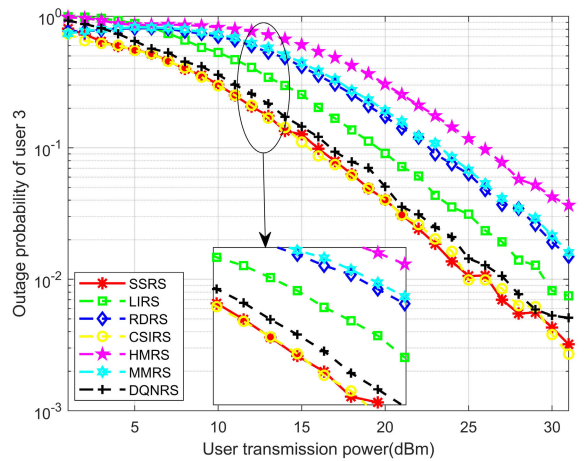


FIGURE 6. Comparison of outage probability of user 3 under different relay selection methods.

is below 10 dBm. This is because a lower power level results in a poorer communication environment; even with relay assistance, some users may fail to connect to a suitable UAV. Once the transmit power exceeds 10 dBm, the fairness curve stabilizes at a high level close to 1. In this regime, all methods can reliably establish relay connections, and our algorithm consistently outperforms the other benchmarks.

Fig. 4, Fig. 5 and Fig. 6 give the outage performance of three pairs of users under six relay selection methods at different user transmit power levels. Among them, when the user's transmit power is less than 5 dBm, the outage probability of each relay selection method is closer, which is because the communication link starts to be established and the signal-to-noise ratio is low, and as the communication gradually starts and the users increase their own transmit power, the outage probability decreases, and the outage performance of the various relay selection methods pulls apart. Comparing the curves in Fig. 4 horizontally, it can be seen that when the outage probability gradually decreases to $10^{-1.5}$ when the user's transmit power required by the six relay

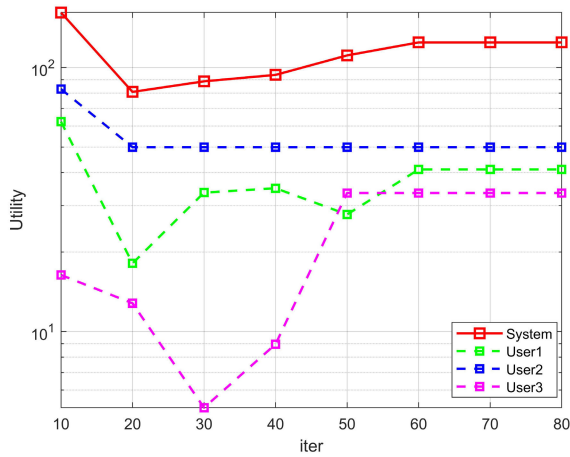


FIGURE 7. The game converges with the number of iterations.

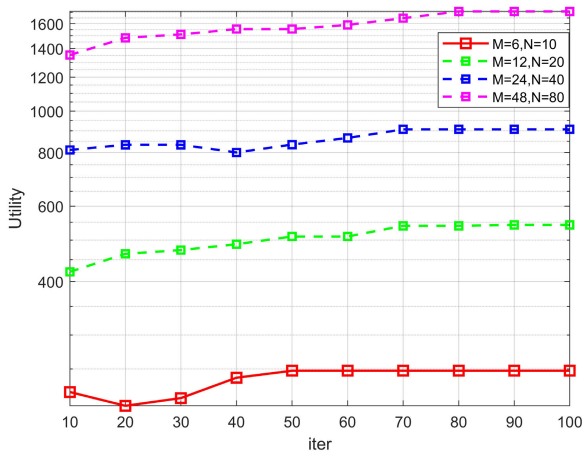


FIGURE 8. Convergence speed under different network scales.

selection methods is greater than 15 dBm, of which the transmit power required by SSRS is the lowest.

A longitudinal comparison of the curves in Fig. 4 shows that when the transmit power is greater than 25 dBm, the probability of outage under each method is lower than 10^{-2} , which can satisfy the need for stability of link communication, and only when the transmit power of the SSRS method does not reach 25 dBm, it can satisfy the need for outage performance.

By analysing Fig. 5, it can be seen that for the outage performance of user 2, the MMRS method has a lower outage probability at the same transmit power, which is because the MMRS algorithm does not take into account the selfishness of the nodes competing for relays, which results in too large a gap in the communication conditions between the three users. Whereas, the SSRS method ensures fairness among the users and avoids creating communication condition gaps to reduce user satisfaction. For user 2, when the transmit power is not higher than 25 dBm, the probability of outage caused by SSRS is also lower than 10^{-2} , which meets the need of communication stability. Fig. 6 gives the plot of outage probability versus transmit power for user three under

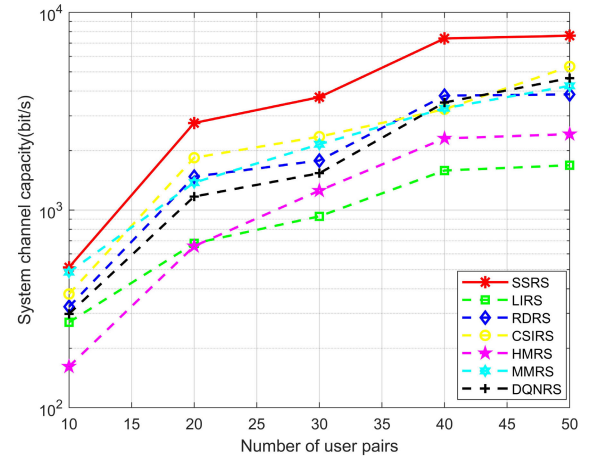


FIGURE 9. System channel capacity varies with the number of user pairs.

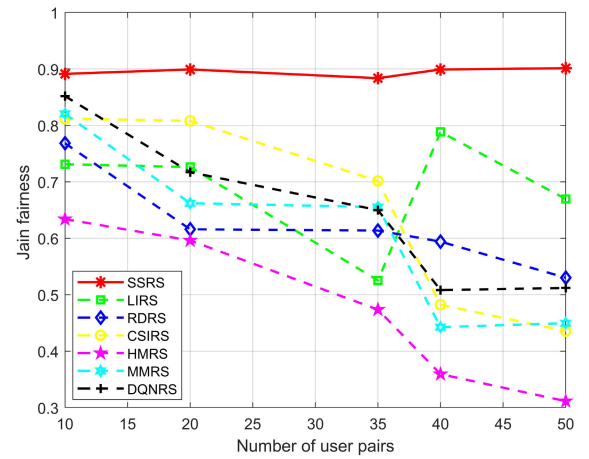


FIGURE 10. System fairness with number of user pairs.

different relay selection methods, from which it can be seen that the SSRS algorithm still has a relatively good outage probability while ensuring user fairness.

The convergence of the proposed game model under the SSRS algorithm is given in Fig. 7. As shown in the figure, the red curve represents the total system utility, the green curve represents the utility of user one, the blue curve represents the utility of user two, and the purple curve represents the utility of user three. By analysing the curves, it can be seen that when the algorithm starts to run, users one and two grab the relay UAVs that are favourable to them and have higher gains, and at this time, the total system gain is also the highest. However, the benefits for User 3 are much lower than those for Users 1 and 2, and there is an excessive difference in communication conditions. When the number of iterations reaches 30, the benefits of user one and user two stabilise, but the benefits of user three reach a minimum. At more than 50 iterations, the gains of the three users gradually regionally stabilise and approach each other, the system gains are also improved over the previous ones, and the algorithm reaches convergence.

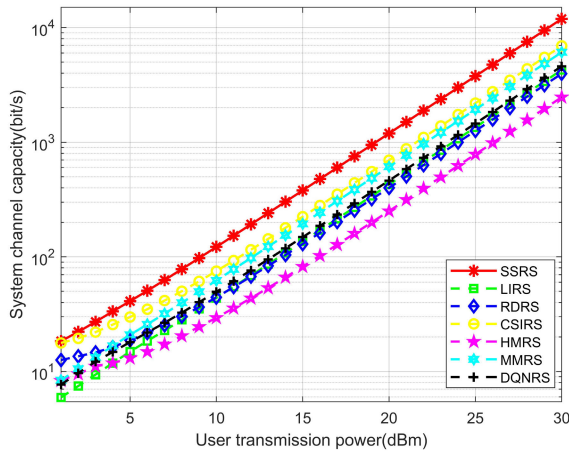


FIGURE 11. Comparison of system capacity under different relay selection methods ($M = 6$ $N = 10$).

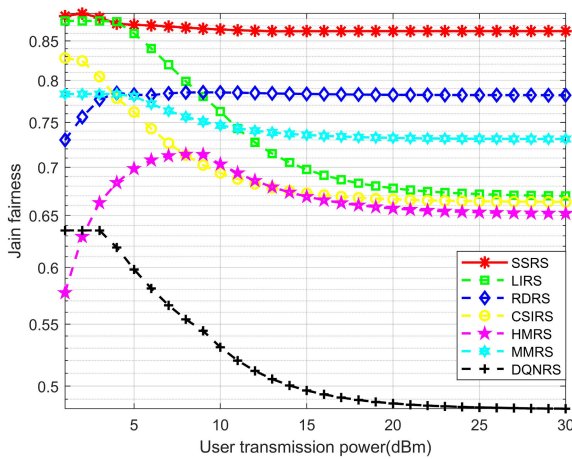


FIGURE 12. Comparison of system fairness under different relay selection methods ($M = 6$ $N = 10$).

Fig. 8 depicts the utility of the proposed algorithm versus the number of iterations under different network scales. The results demonstrate that the algorithm achieves rapid convergence across all four scales. This efficiency is attributed to the diverse relay strategy generation method, which effectively mitigates the risk of becoming trapped in local optima during the convergence process.

Fig. 9 shows the total system capacity versus the number of user pairs, with a transmit power of 20 dBm and 60 relay UAVs. The proposed algorithm consistently surpasses other methods in system capacity. When the number of user pairs is below 20, all curves rise steeply due to abundant relay resources. Beyond this point, the growth rate of system capacity slows down as resources become increasingly contended.

Fig. 10 depicts the variation of system fairness with the number of user pairs, given a transmit power of 20 dBm and 60 relay UAVs. The results demonstrate that the proposed algorithm maintains stable and superior fairness performance compared to other methods across the entire range.

The performance under tight relay resources ($M = 6$, $N = 10$) is shown in Fig. 11 and Fig. 12. An analysis of the curves reveals that the proposed algorithm achieves the best balance between system capacity and user fairness under these constrained conditions.

It is important to note that this study operates under the fundamental premise of relatively abundant relay UAV resources, where the number of UAVs exceeds that of user pairs. Within this context, intelligent relay selection is crucial for maximizing system capacity. Scenarios with scarce or absent relay resources, where communication degenerates to a direct device-to-device (D2D) mode and necessitates a distinct performance analysis paradigm, constitute an independent and valuable direction for future research.

VIII. CONCLUSION

In this paper, we model the user-UAV association in a UAV-assisted cooperative communication network as a network formation game to address the problem of limited system throughput caused by user competition for relay nodes. We correspondingly propose a distributed relay selection algorithm that jointly optimizes system capacity and user fairness. To ensure user fairness, our method employs an alternating iterative process to identify a retained elite strategy set. The relay selection strategy is enhanced through a diversified generation mechanism, which enriches the strategy pool. Simultaneously, we introduce a neighbor machine search replacement method to prevent users from monopolizing relay UAVs with high service quality. Experimental results demonstrate that the proposed algorithm achieves robust performance in both system capacity and user fairness across various network scales. Future work will involve relay UAV trajectory optimization and resource allocation following relay establishment to further enhance service quality. Additionally, integrating reconfigurable intelligent surfaces (RIS) with our algorithm to achieve additional performance gains represents a highly promising research direction.

APPENDIX A PROOF OF THEOREM 2.1

Theorem 2.1 The probability density of a new random variable consisting of the sum of n random variables that are independent and obey different exponential distributions is a linear combination of the density functions of these n exponential distributions.

Calculation of the distribution of independent sums of random variables considering two Gamma distributions. Let $X_1 \sim \Gamma(\alpha_1, \beta_1)$, $X_2 \sim \Gamma(\alpha_2, \beta_2)$. Let X_1 and X_2 be independent of each other. $\alpha_1, \alpha_2 > 0$, $\beta_2 > \beta_1 > 0$.

Let $Y = X_1 + X_2$, Calculate the distribution density of Y . Given the probability density form of $\Gamma(\alpha, \beta)$:

$$\rho_{\alpha, \beta}(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} 1_{(0, \infty)}(x) \quad (41)$$

where $1_{(0, \infty)}(x)$ represents the function's domain of definition from 0 to positive infinity. From the properties

TABLE 2. Summary of notations.

Symbol	Definition
M	Number of source-destination(S-D) pairs
N	Number of relay UAVs
$h_{s;d_i}$	Channel coefficient between the source user i and the destination i
$h_{s;u_j}$	Channel coefficient between the source user i and the UAV j
$h_{u_j;d_j}$	Channel coefficient between the UAV i and the destination j
$d_{s;u_j}$	Distance between the source user i and the UAV j
$d_{u_j;d_j}$	Distance between the UAV i and the destination j
$d_{s;d_i}$	Distance between the source user i and the destination i
P_{u_i}	Power transmitted by the UAV i
x_{u_i}	The first part of the signal uploaded by the source node i
x_{d_i}	The second part of the signal uploaded by the source node i
e	Path loss exponent
B_W	Bandwidth
$n_{s;d_i}$	Background noise
$n_{s;u_j}$	Background noise
$n_{u_j;d_j}$	Background noise
a_1, a_2	Power allocation factor of the two parts of the signals
A, B, C	Defined interrupt events
γ_1, γ_2	Auxiliary variable
γ_3, γ_4	Auxiliary variable
W, Z	Auxiliary variable
$\min$	Minimum function
$\max$	Maximum function
$\log$	Logarithmic function
$\exp$	Exponential function

of probability density and Eq.(41), the probability density of Y is $\rho(x) = \rho_{\alpha_1, \beta_1} * \rho_{\alpha_2, \beta_2}(x)$. $*$ denotes the convolution operation, $\beta_1 \neq \beta_2$, for $x > 0$.

$$\begin{aligned}
& \rho_{\alpha_1, \beta_1} * \rho_{\alpha_2, \beta_2}(x) \\
&= \int_0^x \rho_{\alpha_1, \beta_1}(z) \rho_{\alpha_2, \beta_2}(x-z) dz \\
&= \frac{\beta_1^{\alpha_1} \beta_2^{\alpha_2}}{\Gamma(\alpha_1) \Gamma(\alpha_2)} \int_0^x z^{\alpha_1-1} (x-z)^{\alpha_2-1} e^{-\beta_1 z - \beta_2 (x-z)} dz \\
&\stackrel{z=xt}{=} \frac{\beta_1^{\alpha_1} \beta_2^{\alpha_2}}{\Gamma(\alpha_1) \Gamma(\alpha_2)} x^{\alpha_1+\alpha_2-1} e^{-\beta_2 x} \\
&\quad \times \int_0^1 t^{\alpha_1-1} (1-t)^{\alpha_2-1} e^{(\beta_2-\beta_1)xt} dt. \quad (42)
\end{aligned}$$

An explicit representation of the above integral expression exists when α_1, α_2 are positive integers.

Let $\alpha_1 = \alpha_2 = 1$, $\beta_k = \lambda_k$, $k = 1, 2$, at this time X_1, X_2 obey the exponential distribution of parameters λ_1, λ_2 . The probability density function of $Y = X_1 + X_2$ is:

$$\begin{aligned}
\rho(x) &= \frac{\lambda_2}{\lambda_2 - \lambda_1} \cdot \left[\lambda_1 e^{-\lambda_1 x} \cdot 1_{(0,1)}(x) \right] \\
&\quad - \frac{\lambda_1}{\lambda_2 - \lambda_1} \cdot \left[\lambda_2 e^{-\lambda_2 x} \cdot 1_{(0,1)}(x) \right]. \quad (43)
\end{aligned}$$

This can be thought of as a generalised hybrid exponential distribution. In general, it follows by mathematical induction that if $\{X_k\}_{k=1}^n$ are independent of each other, $X_k \sim \text{Exp}(\lambda_k)$, $k = 1, \dots, n$, $\lambda_1 < \lambda_2 < \dots < \lambda_n$, then the density function of $Y = X_1 + X_2 + \dots + X_n$ is a linear combination of the above exponentially distributed density functions as shown in Eq.(44).

$$\rho = c_1 \cdot \rho_1 + \dots + c_n \cdot \rho_n \quad (44)$$

where, $\rho_k(x) = \lambda_k e^{-\lambda_k x} 1_{(0,\infty)}(x)$, $c_k = \frac{\prod_{j \neq k} \lambda_j}{\prod_{j \neq k} (\lambda_j - \lambda_k)}$, the sum of combination coefficients is 1.

It is worth stating that the portfolio cannot be called a convex portfolio due to the possibility of negative portfolio coefficients, thus distinguishing it from a mixed exponential distribution.

APPENDIX B DESCRIPTION OF THE MAIN MATHEMATICAL SYMBOLS USED

See Table 2.

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